

Ibrahim Hasan Zubairi

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AI engineer who ships. I fine-tune open-source models, build real-time voice agents, and wire AI into messaging APIs, storefronts and internal tools. I use Claude Code, Cursor and MCP servers daily to go from idea to a deployed, tested service fast. Two AI internships plus production systems for clients in Pakistan and South Africa.

EXPERIENCE

AI Intern, CDIS · Systems Limited

Jul 2026 to Present

- Built and deployed an **AI Voice Interviewer**: browser mic audio streamed over WebSockets into the **Gemini Live API** for real-time speech-to-speech, with function-call-driven question flow, live transcription, session resumption, automated scoring and an analytics dashboard.
- Prototyped two other voice stacks for the same product: a Teams meeting bot (**MeetingBaaS**, **Pipecat**, **Deepgram**, **Cartesia**) and a browser interviewer on **ElevenLabs Conversational AI**. Traced failures to Fortinet SSL inspection and blocked QUIC, fixed with an HTTP/2 cloudflared tunnel.
- Wired the interviewer into an automated hiring pipeline (JD generation, intake, screening, interview, offer) with human-in-the-loop review; shipped via a Jenkins CI/CD pipeline to a production VM.
- Building a **Microsoft Teams voice agent** (WASAPI loopback, Gemini Live + TTS, virtual audio routing) that joins and speaks in standups; benchmarked fine-tuned small models against frontier LLMs on cost, latency and accuracy.

Freelance AI & Automation Engineer · Fiverr, Upwork, direct clients

Jan 2025 to Present

- WhatsApp commerce bot** for a South African wellness brand (FastAPI, Meta WhatsApp Cloud API, WooCommerce, Railway): encrypted WhatsApp Flows storefront (RSA/AES-GCM), stateful checkout, and a Google Sheets layer so staff edit every message without redeloys.
- Factory ERP** for a textile manufacturer: barcode-driven Gate Pass Management module live in production on DigitalOcean; production tracking module built and tested.
- SEO and web** for Karachi retailers: cleaned up a 931-product WooCommerce catalog into action lists, fixed redirects and canonicals, schema and LCP issues, optimized Google Business Profiles, and rebuilt a detailing studio's landing page.

AI Engineering Intern (Agentic AI) · Salesflo

Jul 2025 to Aug 2025

- Migrated a WhatsApp AI distributor agent from Make.com to **Google ADK + Vertex AI** (~35% lower latency); multi-agent orchestration over **A2A** with SKU, promotions and ordering tools (+20% decision accuracy, 40% less manual work).

PROJECTS

LoqATS: AI applicant tracking with a fine-tuned local model · FastAPI, Celery, PostgreSQL, Redis, Qwen, Unsloth, Ollama

[github](#)

- Fine-tuned a **Qwen** model with **QLoRA (4-bit) in Unsloth on a 6 GB RTX 3050** to do both resume-to-JSON extraction and resume-vs-JD scoring, exported it and served it through **Ollama** as a local drop-in for the Gemini step, so resumes never leave the machine.
- Multi-tenant SaaS: LLM output constrained to strict JSON feeding a deterministic, explainable scorer; embedding-based skill matching; JWT auth, rate-limited public careers portal. A 55+ case adversarial test pass surfaced six production risks before launch.

Pearls AQI Predictor · LightGBM, Hopsworks, GitHub Actions, FastAPI, Streamlit, SHAP

[github](#)

- Serverless ML system forecasting Karachi's AQI 24/48/72h ahead with hourly ingestion, daily auto-retraining, feature store, model registry and hazard alerts. Six models under walk-forward CV; every deployed model beats persistence (RMSE down 14 to 17%). 169 tests.

PeachyLoopsStudio: AI-assisted content pipeline for my Etsy shop · Python, HTML-to-PDF, SVG

- Built a pipeline that turns a brief into a finished, branded product: 24+ crochet pattern PDFs with code-generated SVG cover art, plus SEO titles, 13-tag sets and bundle strategy for each listing across two product lines.

Agentic outreach pipeline · Claude Code, Apollo MCP, Zoho Mail

- Claude Code as the orchestrator: pulls filtered leads from Apollo and sends from warmed mailboxes with SPF/DKIM/DMARC. Run across UAE, US and Australian real estate.

EDUCATION

BS Computer Science · Karachi School of Business and Leadership · ML, AI, DSA, OS, Linear Algebra

Expected Jul 2027

SKILLS

Open-source models: Qwen, LoRA/QLoRA, Unsloth, Ollama, Hugging Face, embeddings, model evaluation

AI & voice: Gemini API & Live API, Vertex AI, Google ADK, LangChain, Pipecat, ElevenLabs, Cartesia, Deepgram, function calling, A2A

Building with AI: Claude Code, Cursor, MCP servers, agentic workflows

Automation & APIs: Meta WhatsApp Cloud API & Flows, WooCommerce, WordPress, Google Sheets API, Apollo, n8n, Make.com, webhooks

Backend & cloud: Python, FastAPI, Celery, Redis, PostgreSQL, WebSockets, Docker, GCP, DigitalOcean, Railway, Render, Jenkins, GitHub Actions

Security: RSA/AES-GCM payload encryption, JWT auth, rate limiting, SPF/DKIM/DMARC, SSL-inspection debugging